FINAL SUMMARY

1. Consolidated Results Table

Model	Architecture	Parameters	CPU	GPU	TPU
Qwen3-0.6B	Dense	0.596B	✓	✓	✓
Qwen3-0.71B	Dense	0.708B	✓	✓	✓
DeepSeek Sub-1B	MoE	<1B	✓	✓	✓

2. Overall Interpretation

- The experiments show that hardware plays a significant role in transformer training performance. Across all experiments, TPU consistently delivered the fastest execution, followed by GPU and CPU. This is expected because TPUs are specifically designed for large-scale matrix operations, while GPUs provide parallel computation that significantly outperforms CPUs for deep learning workloads.
- The two Qwen models illustrate the effect of scaling a dense transformer. Increasing the model size from 0.6B to approximately 0.71B increased memory usage and computational cost because every additional parameter participates in each forward pass. In contrast, DeepSeek follows a different scaling strategy through its Mixture of Experts architecture. Rather than enlarging every layer, it activates only a subset of experts for each token, resulting in sparse computation while maintaining higher overall model capacity.
- Overall, these experiments demonstrate that model size, architecture, and hardware all influence training performance. Dense transformers scale by increasing the number of active parameters, whereas MoE models improve scalability through expert routing. Running the experiments on CPU, GPU and TPU also highlighted how specialized hardware can significantly reduce training time without changing the underlying model configuration.

3. Unexpected Findings

- The synthetic dataset caused the training loss to decrease rapidly because it contains repetitive patterns rather than meaningful language, making it suitable for benchmarking but not for evaluating model quality.
- Scaling Qwen from 0.6B to 0.71B required only a few configuration changes, but the increase in memory usage was clearly noticeable across all backends.
- The DeepSeek MoE model required additional configuration compared to the dense Qwen models due to expert routing and MoE-specific parameters, making the setup process more involved.
- Despite the architectural differences, the same training workflow and synthetic dataset could be used successfully across CPU, GPU and TPU, allowing for a consistent comparison of execution behavior.